DEMON: Dynamic Exploration of Molecular Simulations Through Optimized Navigation

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Introduction

Molecular dynamics (MD) simulations use physics to model atomic motion, revealing processes such as protein conformational changes and drug binding. However, biologically important transitions often require crossing high free-energy barriers and occur far beyond the timescales accessible to conventional simulation.[1]

Enhanced-sampling methods accelerate exploration but often depend on predefined reaction coordinates or other system-specific features.[2] DEMON (Dynamic Exploration of Molecular Simulations through Optimized Navigation) instead uses reinforcement learning to allocate computational effort across an ensemble of trajectories. At each decision point, the agent selects one trajectory to replicate while another is pruned, maintaining a fixed ensemble size. By learning directly from atomic trajectories, DEMON aims to develop transferable exploration strategies across molecular systems.

DEMON as a Split-Prune Decision Environment

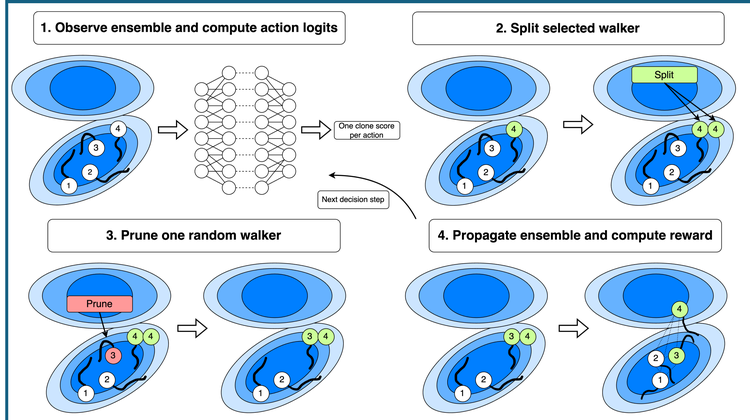


Figure 1: DEMON formulates ensemble sampling as a reinforcement learning problem. The agent repeatedly reallocates a fixed simulation budget toward selected trajectories, using the resulting diversity reward to improve future decisions.

Learning to Allocate Simulation Effort

DEMON learns which replica to clone so that a fixed simulation budget ensemble explores distinct configurations more effectively. At each decision step, a discrete Soft Actor–Critic[3] policy scores the current ensemble, selects one replica to replicate, and learns from the resulting diversity of the propagated ensemble.

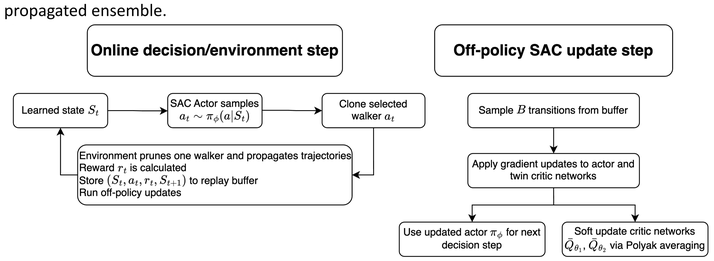


Figure 2: Off-policy control of a fixed MD budget. Replay reuses costly MD segments, while random pruning holds ensemble size constant.

DEMON is rewarded when propagated replicas occupy more diverse configurations, encouraging exploration of different states.

$$r_t \propto -\frac{2}{N(N-1)} \sum_{i<j} \exp\left[-\frac{d(x_i, x_j)}{\sigma}\right]$$

Where N represents the number of walkers and $d(x_i, x_j)$ represents the Euclidean or dihedral space distance between walkers, and σ is a reward scale/saturation parameter.

Representation Learning Architecture

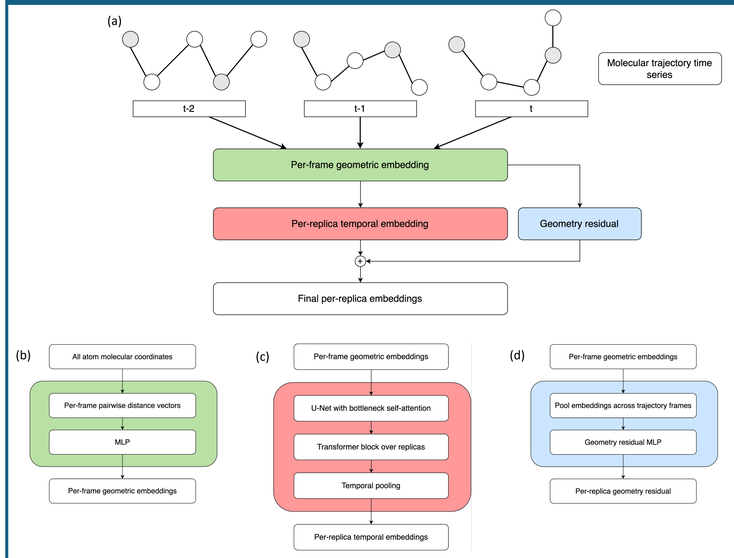


Figure 3: Coordinate histories are converted into decision-ready embeddings for each replica. Per-frame geometry is encoded with an MLP; a U-Net–based temporal encoder[4] processes trajectory history, and a Transformer block[5] exchanges context across replicas. These features are combined with a geometry residual to produce the SAC state representation.

Model System Results

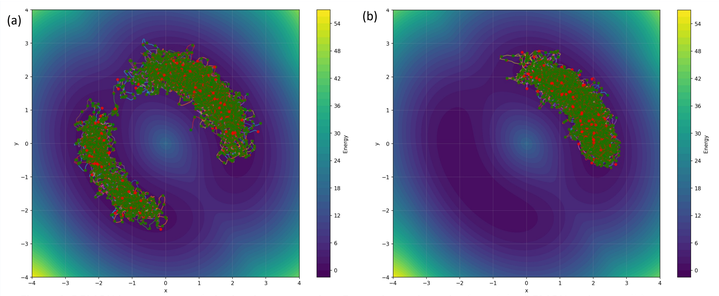


Figure 4: DEMON improves out of distribution exploration. Evaluation trajectories from **(a)** a DEMON policy and **(b)** random splitting, both using 20 walkers over 200 decision steps. On a landscape rotated to $\frac{4\pi}{5}$, outside of the $\left[0, \frac{\pi}{2}\right]$ training distribution, DEMON explores both basins while random splitting remains in the initial basin. The DEMON policy was trained for 50,000 decision steps, using a stochastic environment reset rate of 0.005; when the environment was reset walkers were reinitialized in the starting basin and the environment was rotated to a new random angle. Model system trajectories are propagated using overdamped Langevin dynamics.

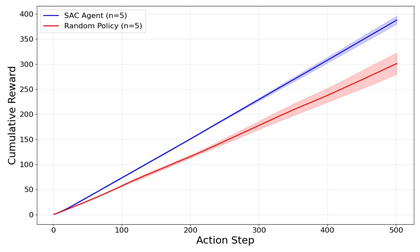


Figure 5: DEMON achieves higher cumulative exploration reward (mean $\pm$1SD; $n = 5$) than random replica selection on an OOD 8-dimensional multi-basin GMM landscape. Evaluated over 500 action steps after 100,000 training steps. The randomly chosen evaluation seed dictates basin placement, size, and shape. All walkers are initialized within the deepest basin for both DEMON and random evaluations.

As shown, the DEMON agent generalizes without fine-tuning to out-of-distribution model landscapes. Evaluation landscapes share the dimensionality and broad structure of the training distribution but use rotation angles and/or random seeds withheld during training. DEMON's performance against random splitting on out-of-distribution environments indicates that it learns transferable features of ensemble exploration rather than memorizing individual landscape orientations or realizations. These controlled results provide a foundation for extending learned allocation strategies to more complex molecular systems.

Biomolecular Results

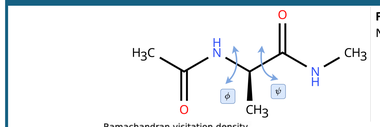


Figure 6: Capped alanine dipeptide structure (Ace-Ala-Nme; Ala2) with labeled ϕ and ψ dihedral angles.

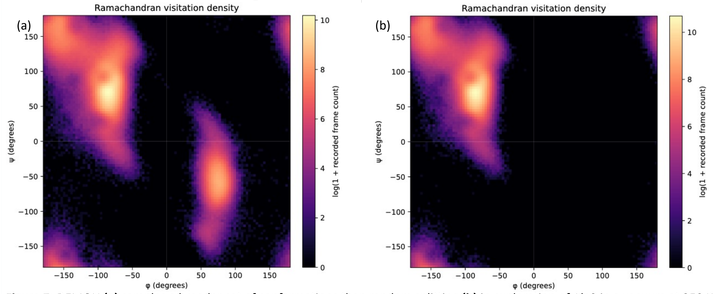


Figure 7: DEMON **(a)** samples a broader set of conformations than random splitting **(b)** in exploration of Ala2 in a vacuum at 350 K. DEMON explores the C_{7eq}, C_{7ax}, and C_5 conformations, while random splitting only explores C_{7eq} and C_5 conformations. Both evaluations consisted of 2,000 decision steps with an 8 ps decision interval, totaling 192 ns across 12 walkers. The DEMON agent was trained for 10 episodes of 4,000 decision steps, 3.84 µs of total MD simulation time.

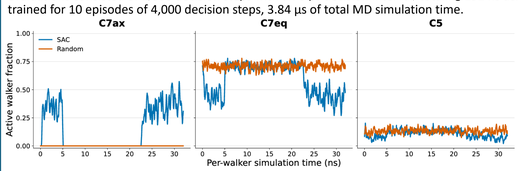


Figure 8: Fraction of active walkers in each basins over per-walker simulation time. Lines show 25-action rolling means for DEMON (blue) and random selection (orange). Each evaluation ran for 4,000 decision steps of 8 ps per decision, totaling 32 ns per walker. SAC intermittently maintains walkers in C7ax, whereas random selection does not.

Conclusions & Future Work

DEMON improves exploration across model systems. Across 2D and 8D synthetic landscapes, DEMON consistently outperforms random splitting and transfers zero-shot to unseen landscape configurations, including held-out rotations and newly seeded landscapes.

Early atomistic results are positive. On alanine dipeptide (Ala2) in vacuum, early DEMON agents achieve greater configurational diversity than random splitting. Robust performance at the target operating temperature of 300 K remains a challenge.

Address rare events during training. At lower temperatures, rare transitions occur too infrequently for the agent to learn effective allocation. A temperature curriculum—beginning at higher temperature and annealing toward 300 K—may provide useful early exploration.

Develop a scalable molecular representation. The current representation architecture is still being refined and will need to scale efficiently to larger peptides and proteins.

Aim for transferability on unseen peptide systems. The next core goal of DEMON is to learn a policy that transfers across small OOD polypeptide systems and reproduces accurate thermodynamic and kinetic quantities.

References

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